

Flow Matching Predictive Control with Constraints

Master's Thesis in Informatics · Defence

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DPCC: projection inside a fixed denoising sequence

- Diffusion planners capture multimodal demonstrations, and Diffusion Predictive Control with Constraints (DPCC) projects their denoising iterates onto the feasible set.
- In the inherited implementation the denoising sequence is fixed at training and repeated at every control step.
- Every control step therefore pays for the whole sequence, and the projection with it.

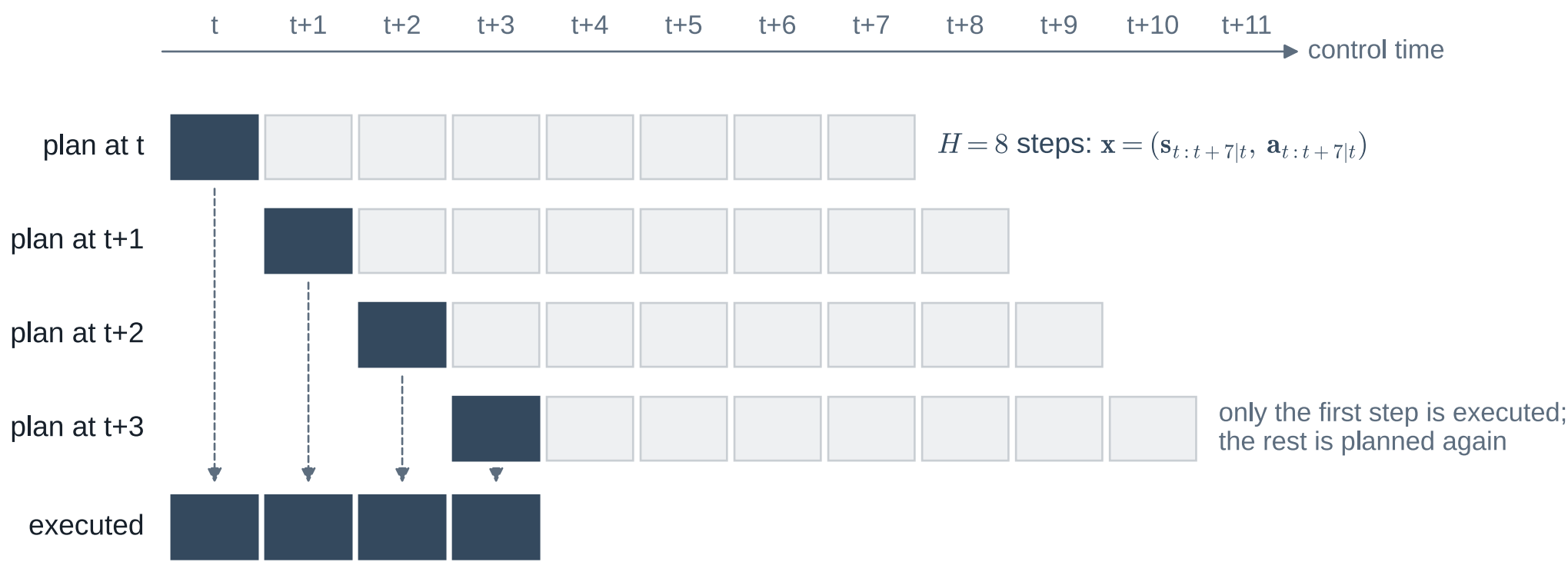
Idea and research questions

This thesis replaces diffusion with deterministic transport along an ordinary differential equation, whose step budget is chosen at deployment.

- **RQ1 — Generative model.** Can the flow-based models reach the diffusion baseline's task success and constraint satisfaction with fewer sampling steps and less planning time per control step?
- **RQ2 — Constraints.** Does endpoint projection reach the goal with fewer constraint violations, fewer control steps or less computation than the per-step projection of DPCC?
- **RQ3 — Transfer.** Do the answers to RQ1 and RQ2 hold when the observation becomes camera images, when the task requires aligning an object through contact, and when the platform is a quadrotor?

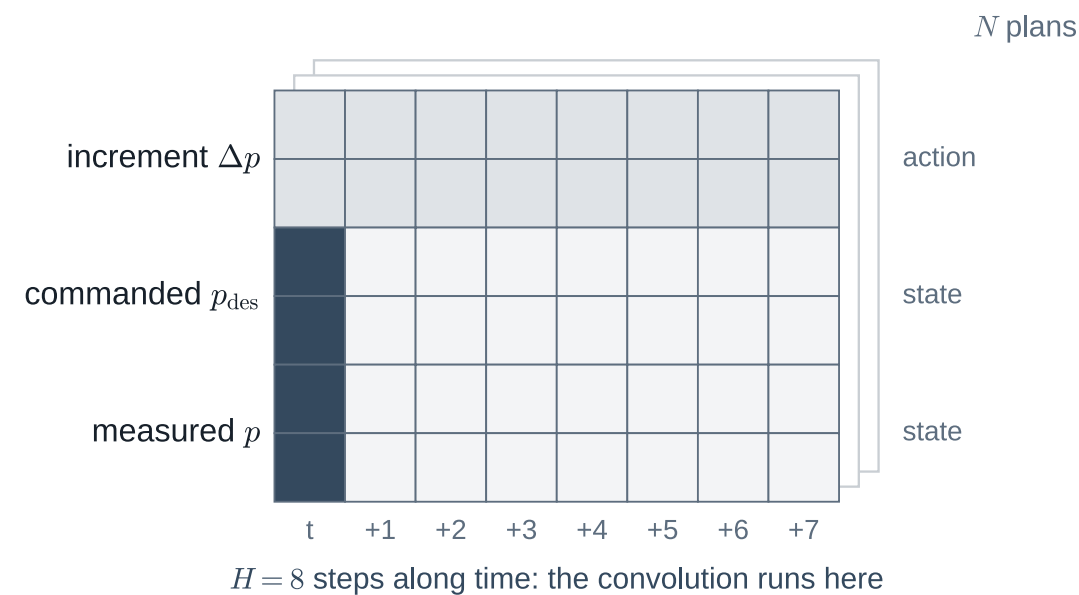
Three steps: **D3IL-avoiding**, the benchmark of DPCC unchanged → **D3IL-aligning**, observed through cameras → **quadrotor**, three scenes.

Receding horizon: plan eight steps, execute the first, plan again



Inside one control step. The generative objective and the projection are the two varied components.

The network reads a plan as an $8 \times d$ signal



The plan tensor of D3IL-avoiding, $(N, 8, 6)$. Dark: the known current state, imposed by inpainting and masked in the loss.

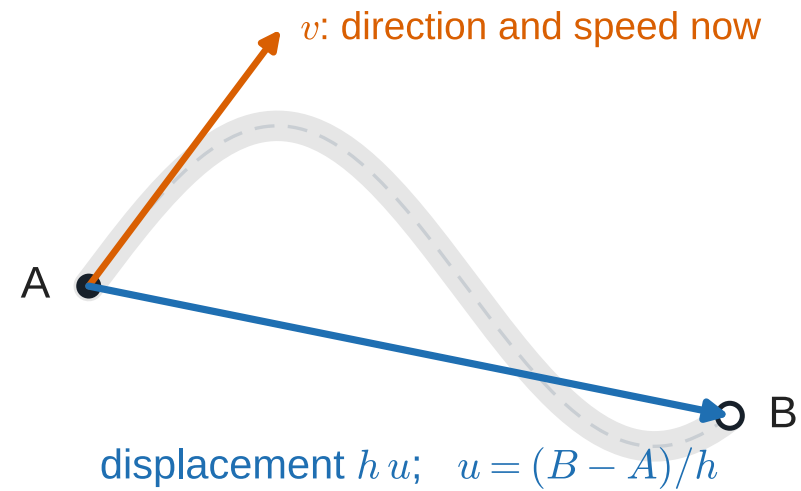
Environment	Plan in / out	State anchor
D3IL-avoiding	$(N, 8, 6)$	$(N, 4)$
D3IL-aligning	$(N, 8, 9)$	$(N, 6)$
UAV-corridor, UAV-s-curve	$(N, 8, 12)$	$(N, 9)$

D3IL-aligning adds two RGB views, $2 \times (N, 1, 3, 96, 96)$, encoded to $(N, 128)$. UAV-pillars flies the D3IL-avoiding models.

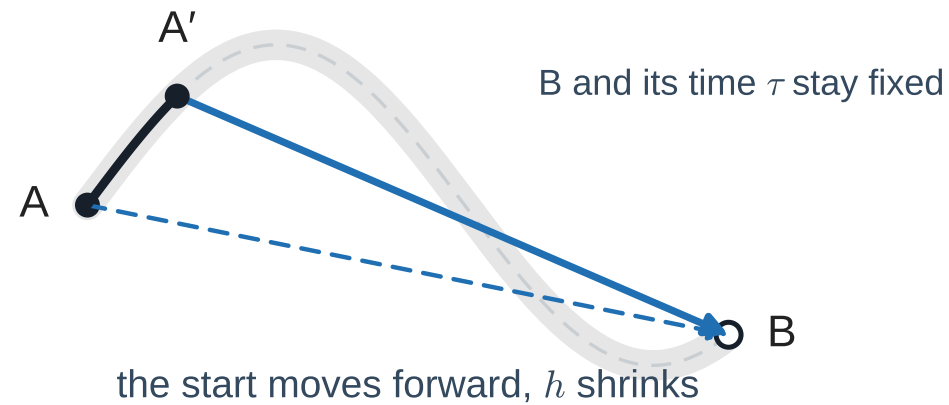
- A temporal U-Net: convolution along time, channel widths $(d, w, 2w, 4w, 8w)$, three stride-two stages ($8 \rightarrow 4 \rightarrow 2 \rightarrow 1$) with skip connections.
- Conditioning is a per-channel bias in every block: the time embedding of τ , plus h for MeanFM and CI-MeanFM, plus the image latent on D3IL-aligning.
- The output has the plan's shape: v_θ , u_θ , or the baseline's noise ϵ_θ .

A car on a winding road: what the average-velocity objectives learn

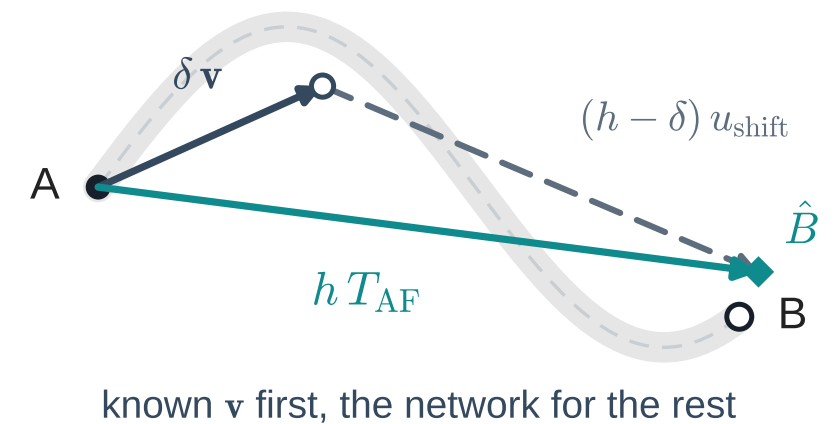
1 · velocity and average velocity



2 · MeanFM: move the start, keep the destination



3 · CI-MeanFM: a target in two parts



$$h u(\mathbf{x}_r, r, h) = \mathbf{x}_{r+h} - \mathbf{x}_r$$

one step from the start:

$$\hat{\mathbf{x}}_1 = \mathbf{x}_0 + u_\theta(\mathbf{x}_0, 0, 1)$$

$$u = v + h \frac{d}{dr} u$$

target $\text{sg}(v + h \text{JVP})$: a derivative at the current input

$$h u_{\text{tgt}} = \delta v + (h - \delta) u_{\text{shift}}$$

$\delta = \alpha h$; a prediction at a shifted input; α from 1 to α_{end} in training

The road stands for a trajectory through the space of candidate plans; h is generation time, not seconds. A drawing, not thesis data.

Three objectives inside the same constrained controller

FM

instantaneous-velocity matching

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E} \left\| v_{\theta}((1 - \tau)\mathbf{x}_0 + \tau\mathbf{x}_1, \tau, \mathbf{c}) - (\mathbf{x}_1 - \mathbf{x}_0) \right\|_2^2$$

MeanFM

analytic average-velocity matching

$$u_{\text{tgt}} = \text{sg} \left[\mathbf{v} + h \text{JVP} \left(u_{\theta}; (\mathbf{x}_r, r, h), (\mathbf{v}, +1, -1) \right) \right]$$

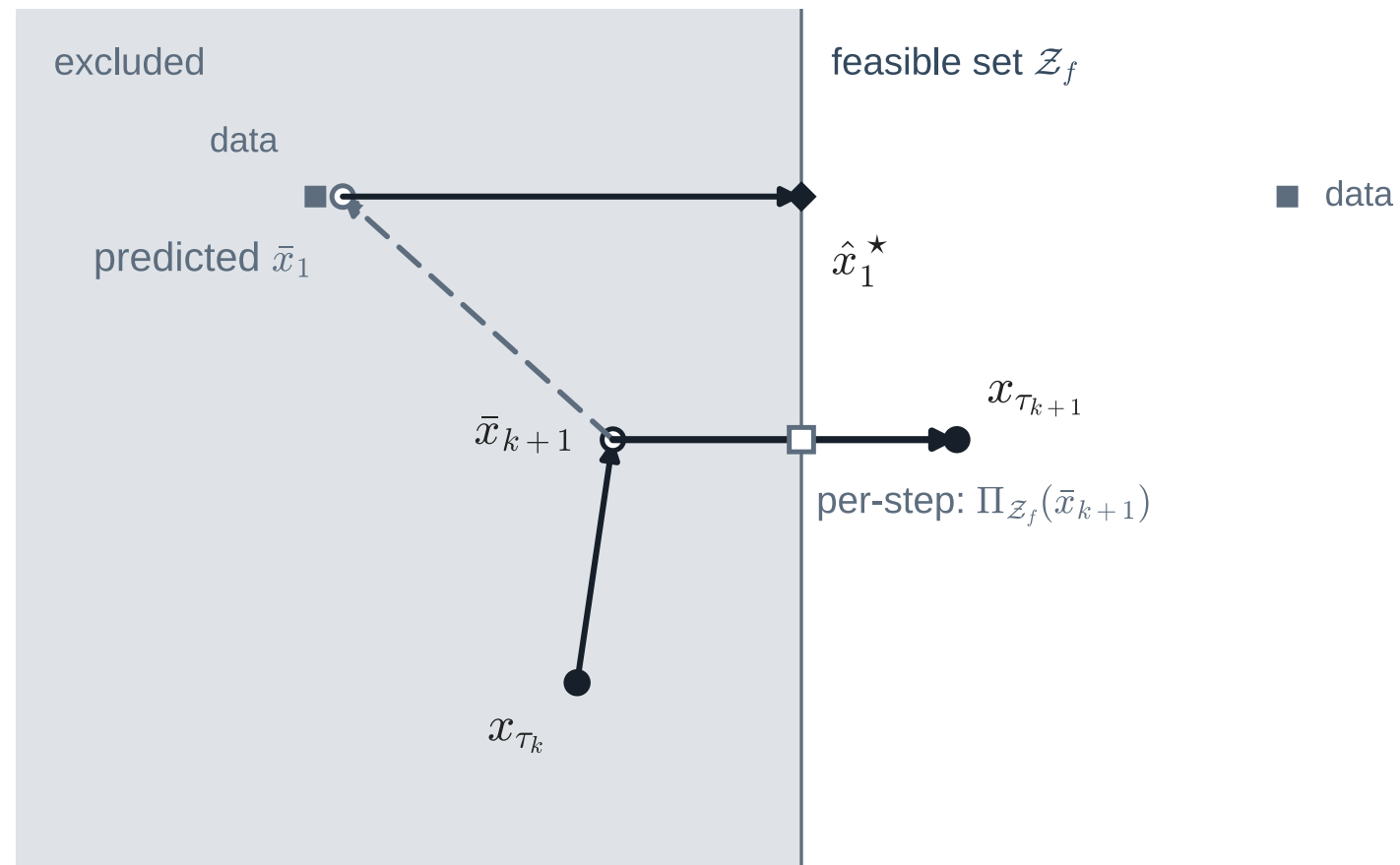
CI-MeanFM

consistency-interpolated average-velocity
matching

$$u_{\text{tgt}} = \alpha \mathbf{v} + (1 - \alpha) \text{sg} \left(u_{\theta}(\mathbf{x}_r + \delta \mathbf{v}, r + \delta, h - \delta) \right), \quad \delta = \alpha h$$

$\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_0$; sg: stop-gradient. Backbone, projection, data and evaluation protocol are kept; the number of network evaluations becomes a choice made at deployment.

Per-step projection constrains the current iterate, endpoint projection a predicted final plan



One guided step of endpoint projection:

$$\bar{\mathbf{x}}_{k+1} = \mathbf{x}_{\tau_k} + \Delta\tau v_\theta(\mathbf{x}_{\tau_k}, \tau_k, \mathbf{c})$$

$$\bar{\mathbf{x}}_1 = \bar{\mathbf{x}}_{k+1} + (1 - \tau_{k+1}) v_\theta(\bar{\mathbf{x}}_{k+1}, \tau_{k+1}, \mathbf{c})$$

$$\hat{\mathbf{x}}_1^\star = \arg \min_{\hat{\mathbf{x}}_1 \in \mathcal{Z}_f} \|\hat{\mathbf{x}}_1 - \bar{\mathbf{x}}_1\|_2^2$$

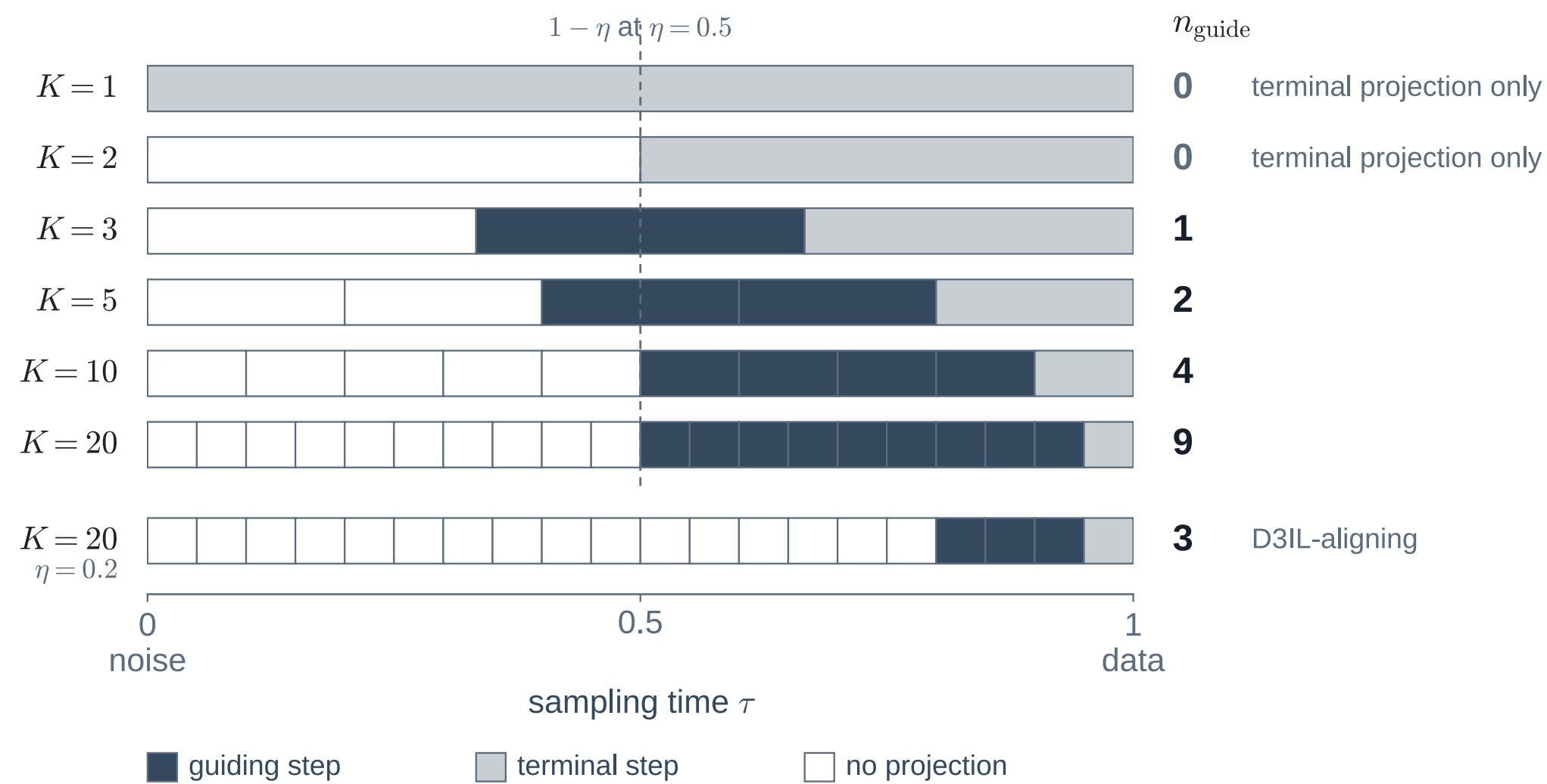
$$\mathbf{x}_{\tau_{k+1}} = \bar{\mathbf{x}}_{k+1} + \tau_{k+1} (\hat{\mathbf{x}}_1^\star - \bar{\mathbf{x}}_1)$$

Per-step projection (DPCC) projects the iterate itself:

$$\mathbf{x}_{\tau_{k+1}} = \Pi_{\mathcal{Z}_f}(\bar{\mathbf{x}}_{k+1}).$$

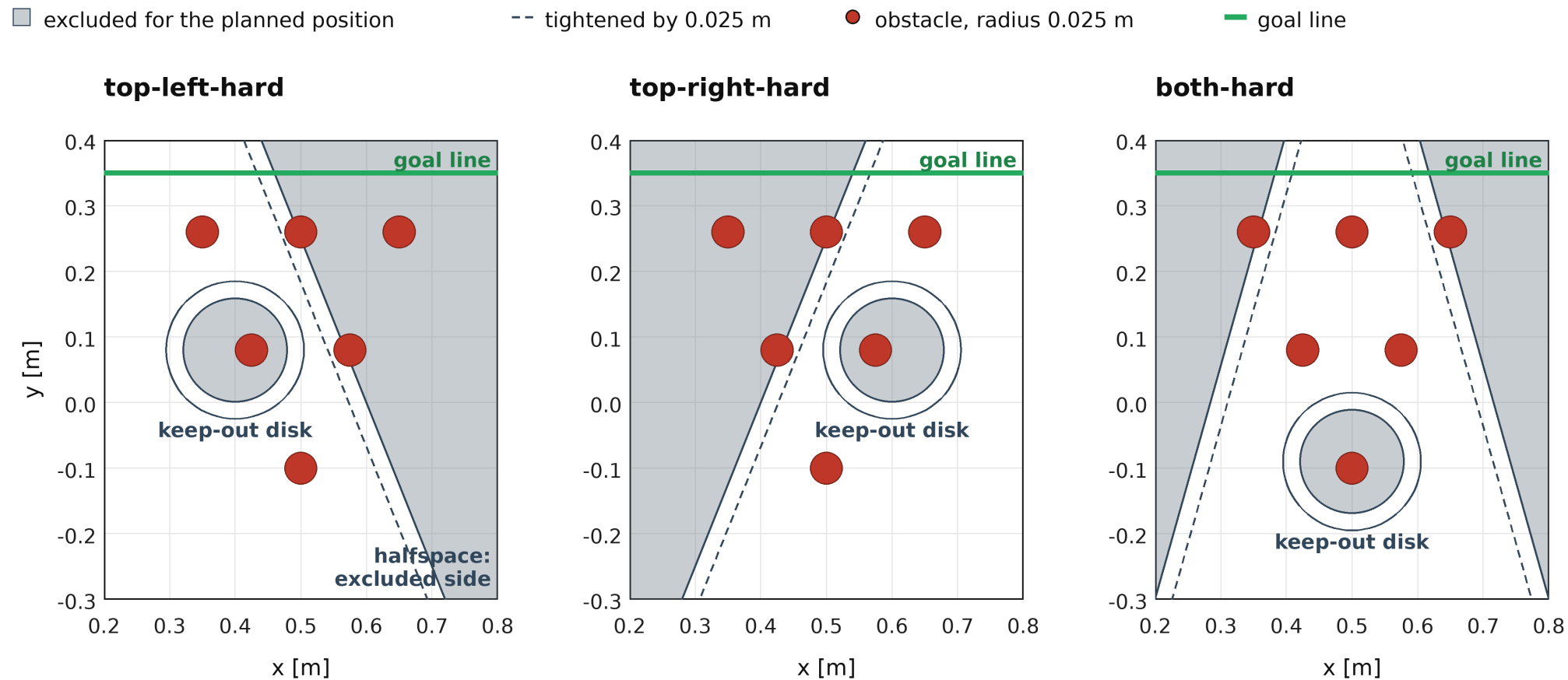
Toy: two data points, a halfspace, one step at $\tau_k = 0.5$, $K = 4$; computed, not thesis data.

Endpoint projection guides only where the budget leaves a step before the last



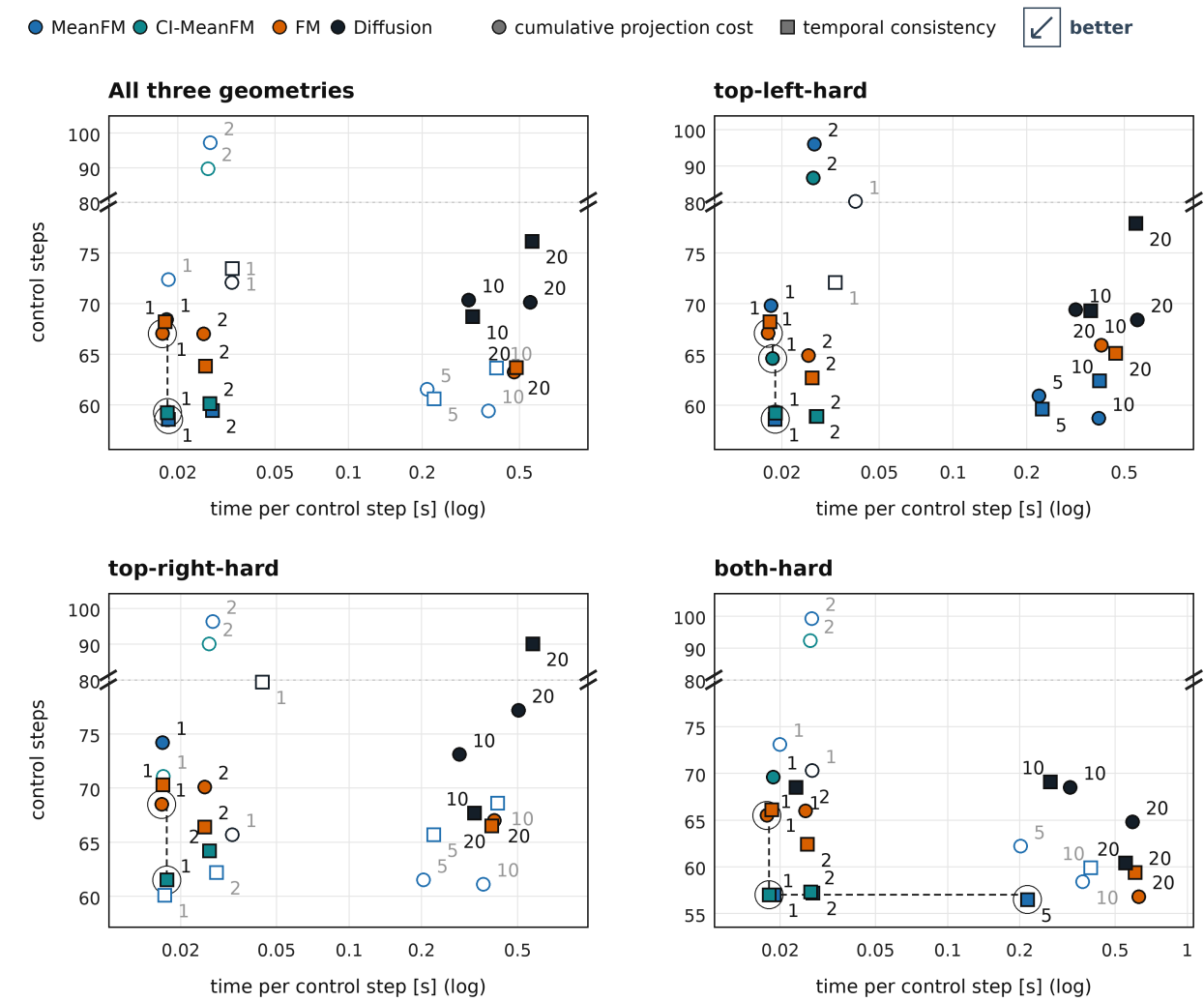
$n_{\text{guide}}(K, \eta) = \lceil \eta K \rceil - 1$ for $\eta > 0$: the active steps are the last $\lceil \eta K \rceil$, and the terminal one does not guide.

D3IL-avoiding: the benchmark of DPCC, unchanged

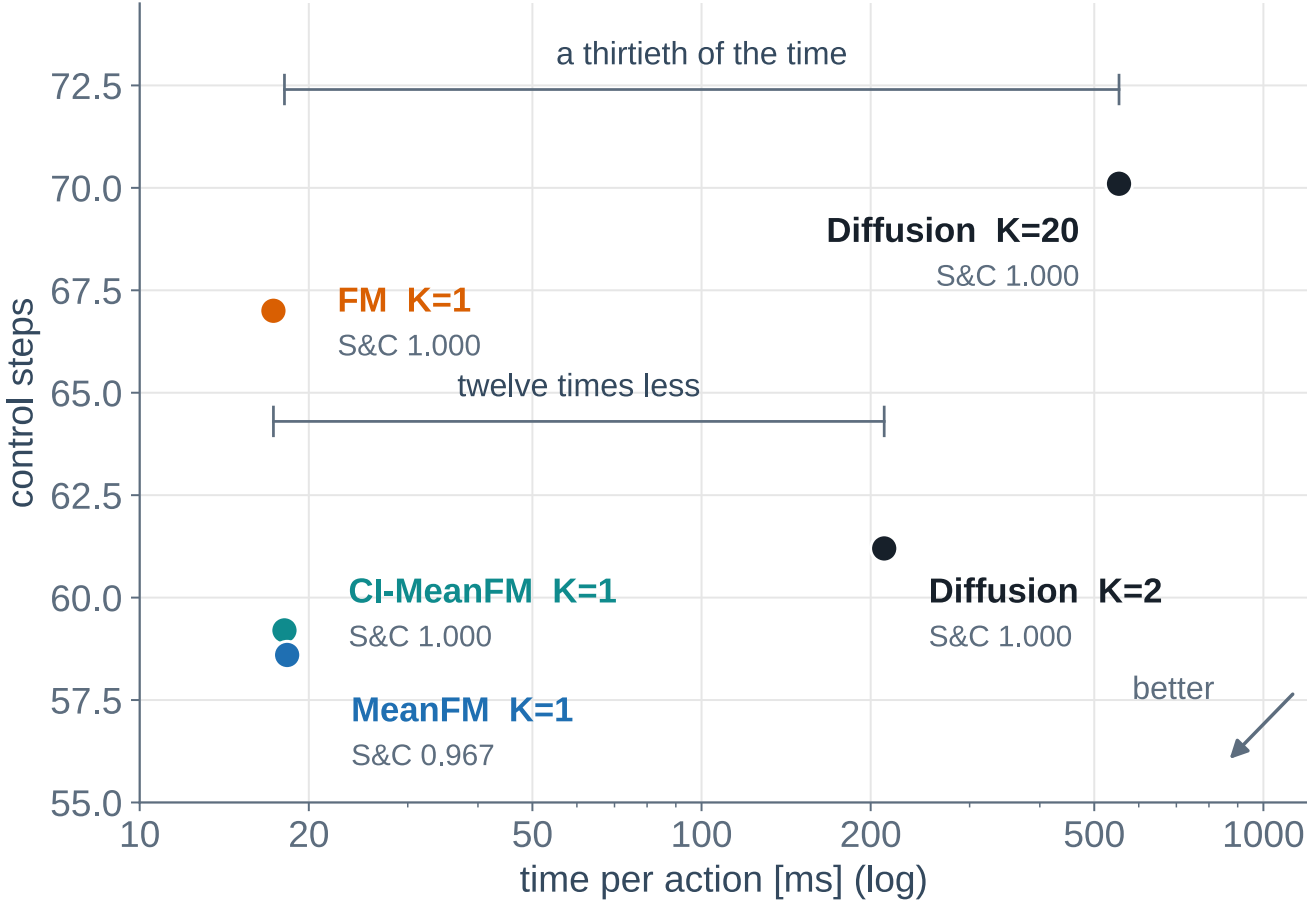


- The Panda manipulator moves its end effector across a field of six obstacles to a goal line.
- Three constraint geometries; the same environment, demonstrations, projection and protocol as DPCC: five training seeds × two episodes per geometry.

On the benchmark of DPCC, the average-velocity models dominate its diffusion model



Every budget and rule (thesis figure).



The operating points. Five training seeds $\times$ two episodes per geometry, tightened constraints, temporal U-Net of 4.0 M parameters.

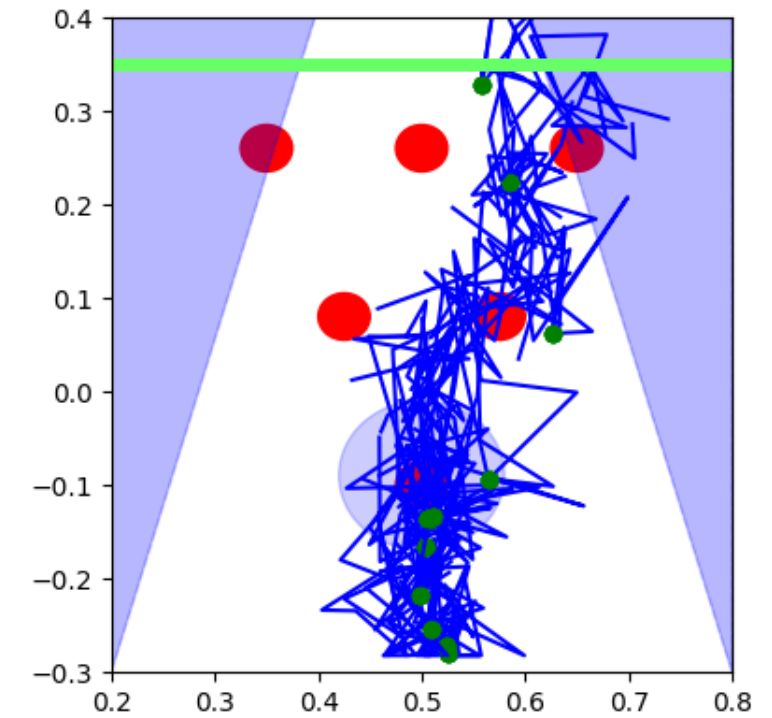
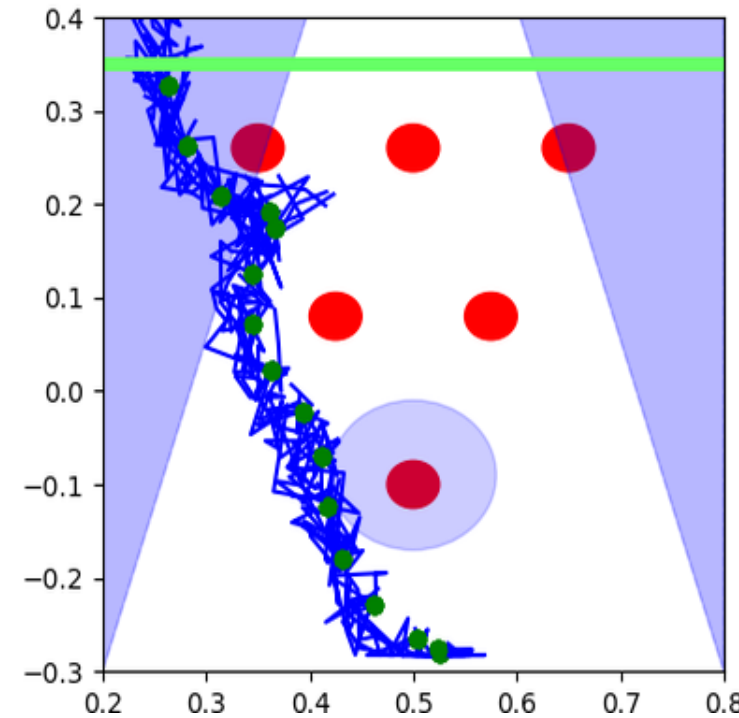
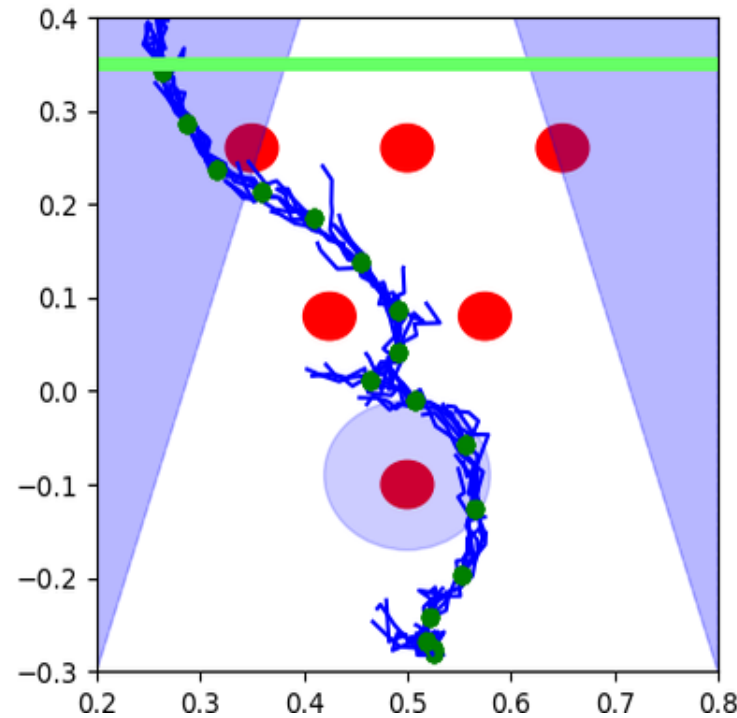
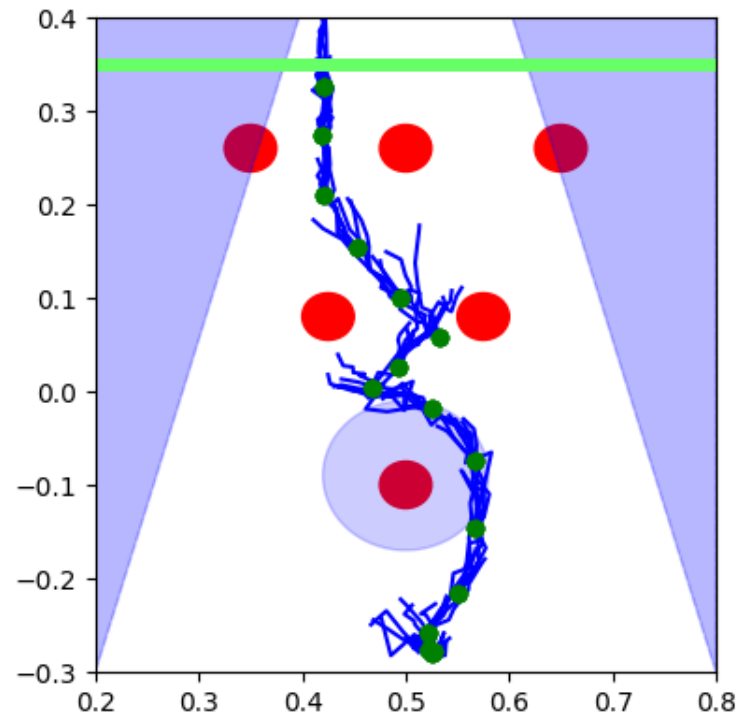
The time a projection takes depends on how far from feasible the plan it receives is

MeanFM

CI-MeanFM

FM

Diffusion



Plans without projection at $K = 1$, every fourth control step, all four candidates; both-hard, one training seed. The executed path is not drawn.

- At one evaluation MeanFM's successive plans form a narrow ribbon; the diffusion baseline's plans jump across the workspace from one control step to the next.

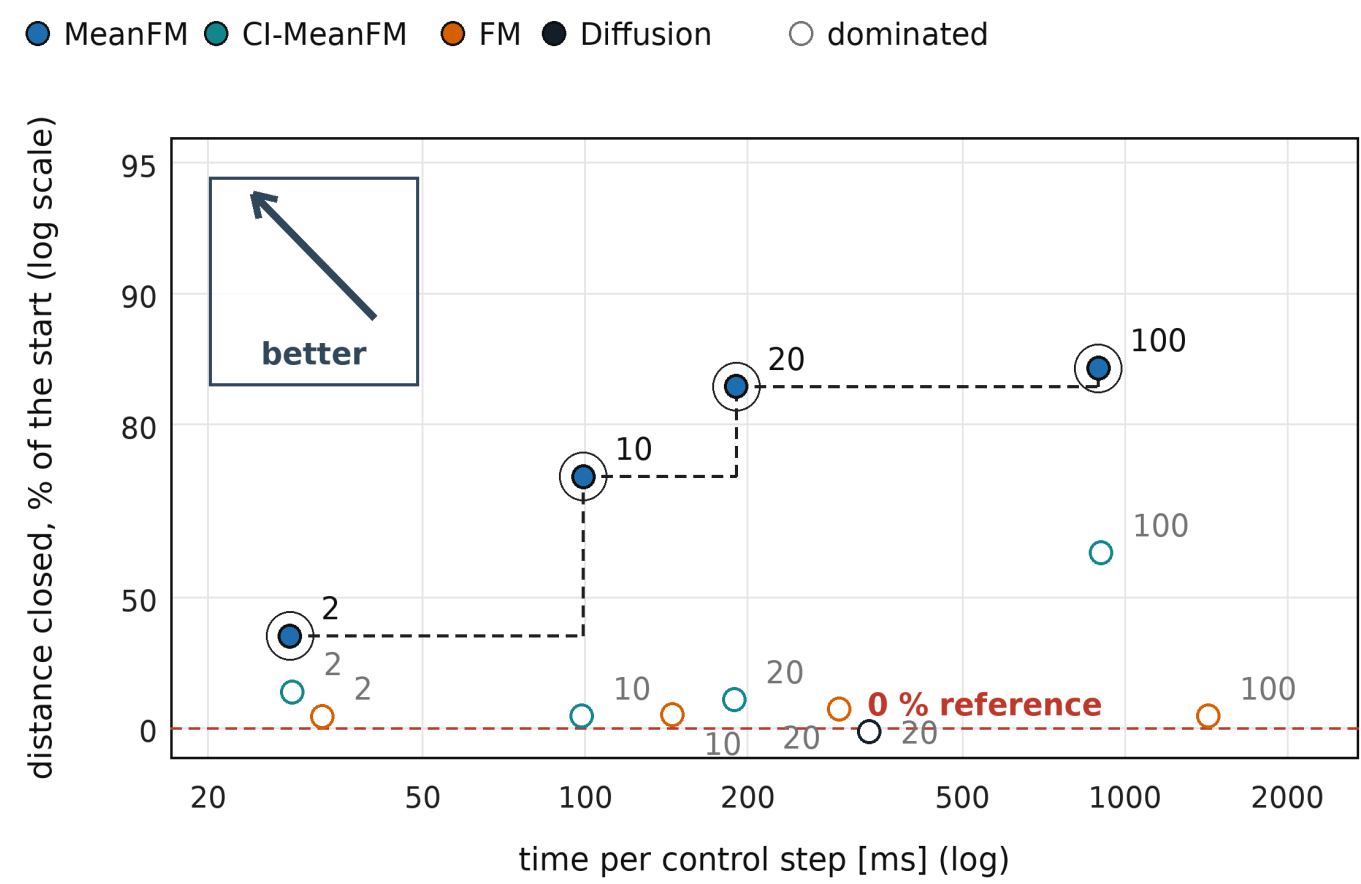
D3IL-aligning: the robot observes two camera views



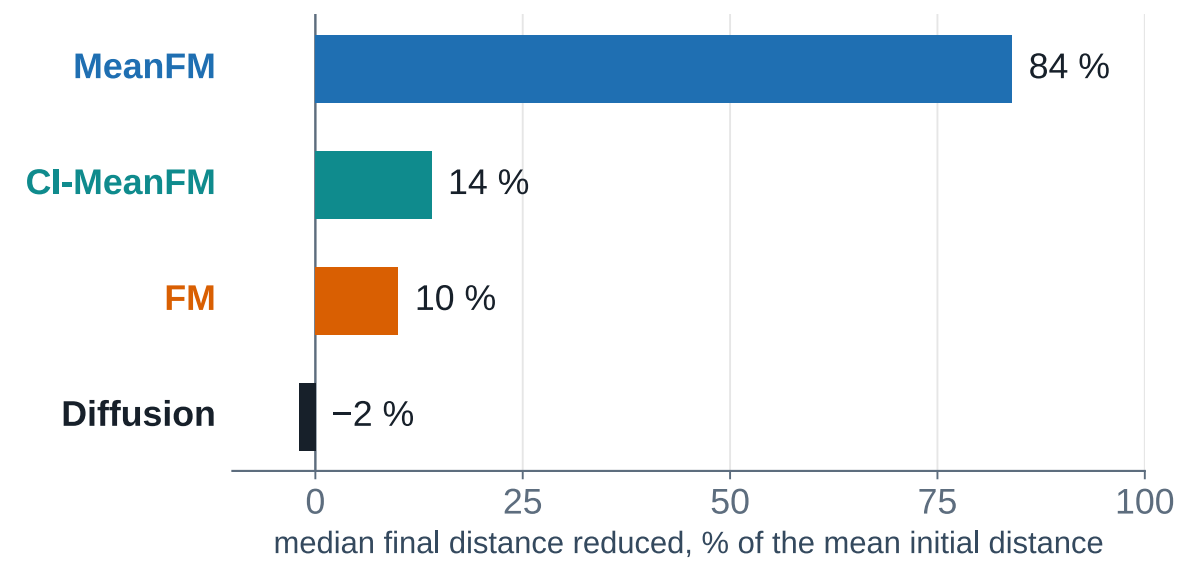
MuJoCo demo pending. Placeholder: a random development rollout from temp/, not a thesis result.

- The manipulator pushes a box into a target pose; each episode starts from a context, an initial box pose and a target pose.
- The observation carries no box or target pose: the policy must locate the box in two RGB streams, a front view and an in-hand view.
- The generative network is given a visual encoder; the projection is given the dynamics rows and the constraint set of the new task.

On D3IL-aligning the dominance repeats where the task separates the models



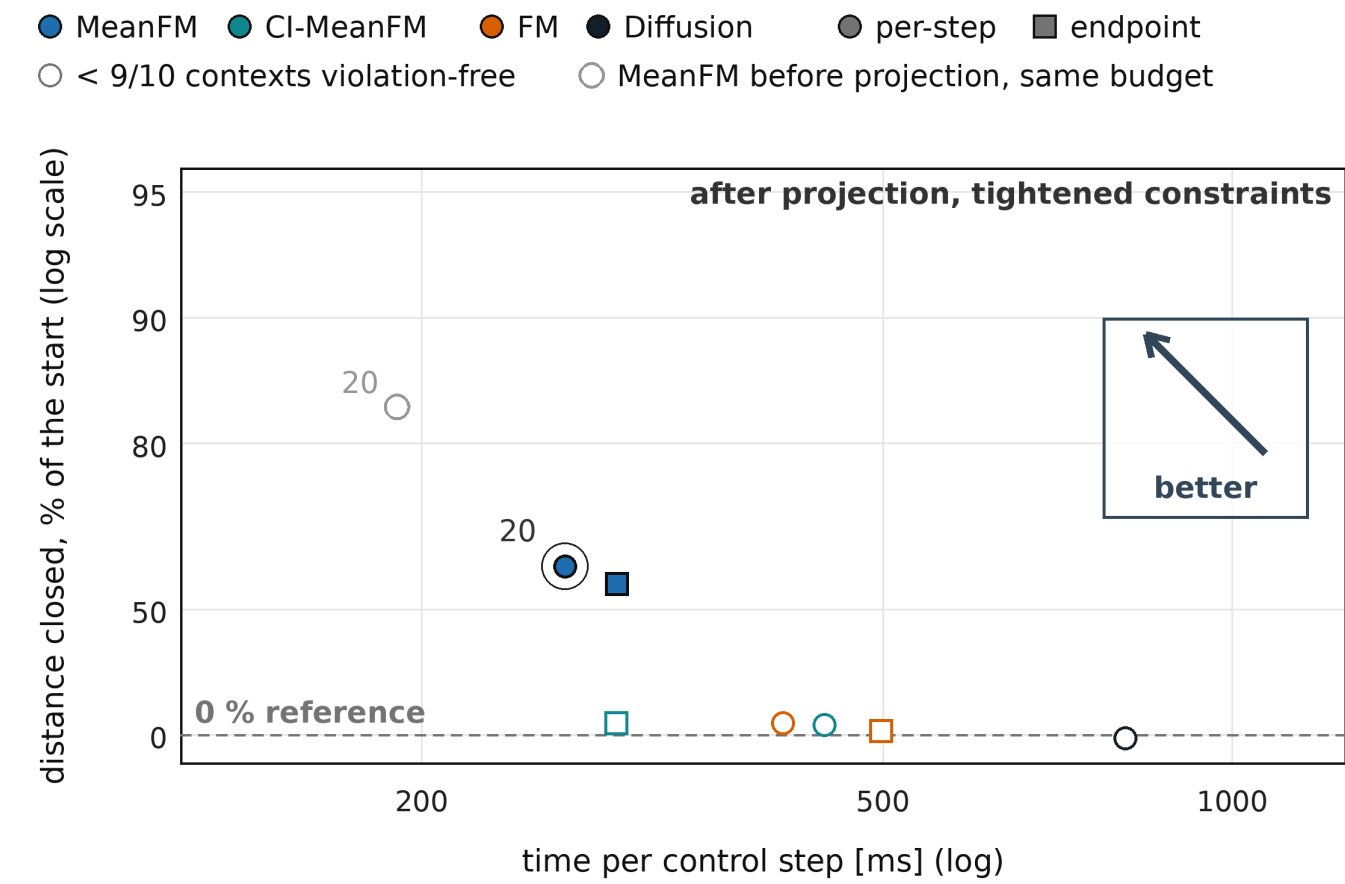
Distance closed against time per control step, every budget (thesis figure).



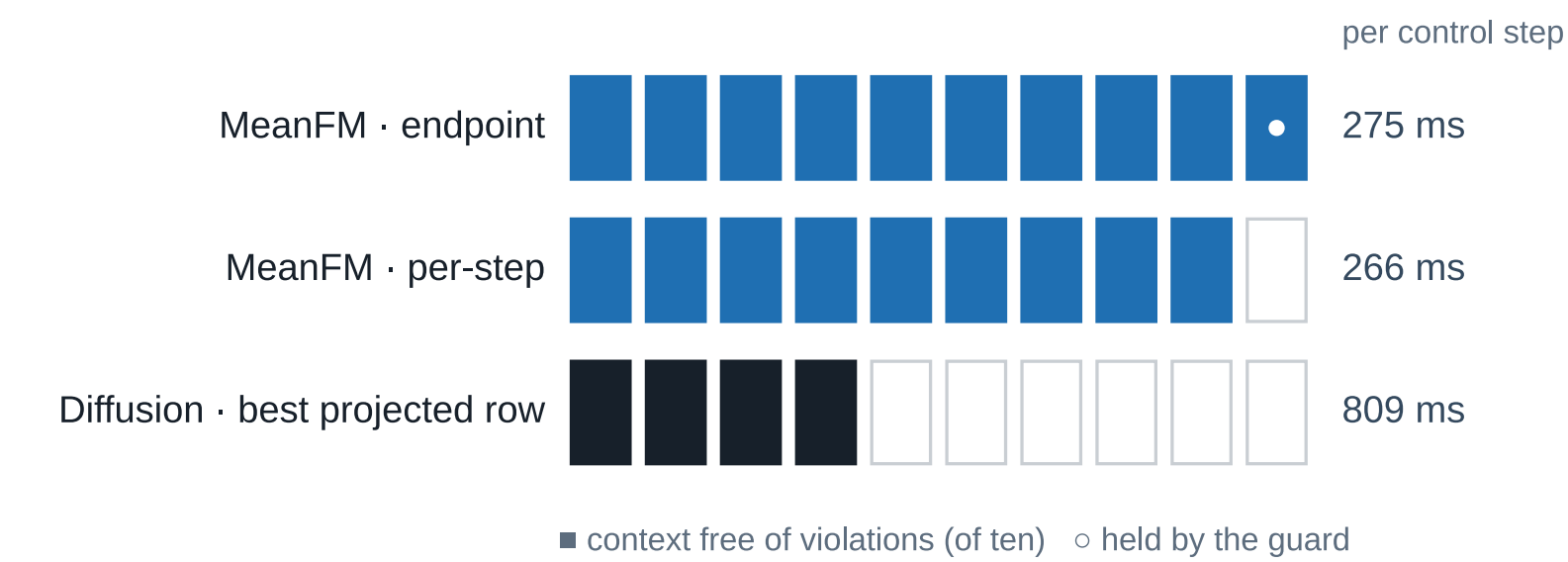
Without projection at $K = 20$, ten training-set contexts.

- At two evaluations, at a seventh to a tenth of their cost, MeanFM's reduction is already two fifths.
- **CI-MeanFM** does not improve with the budget below a hundred evaluations.

At the budget this task needs, endpoint projection is the better projector on the constraint



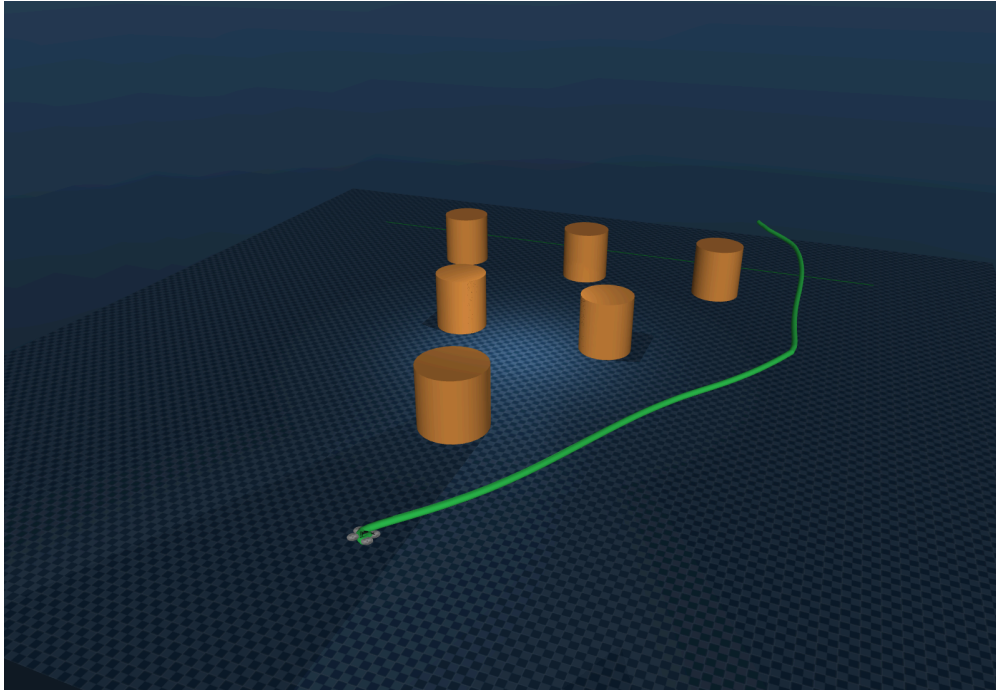
After projection, tightened constraints; circles per-step, squares endpoint (thesis figure).



Contexts free of violations after projection, $K = 20$, random selection.

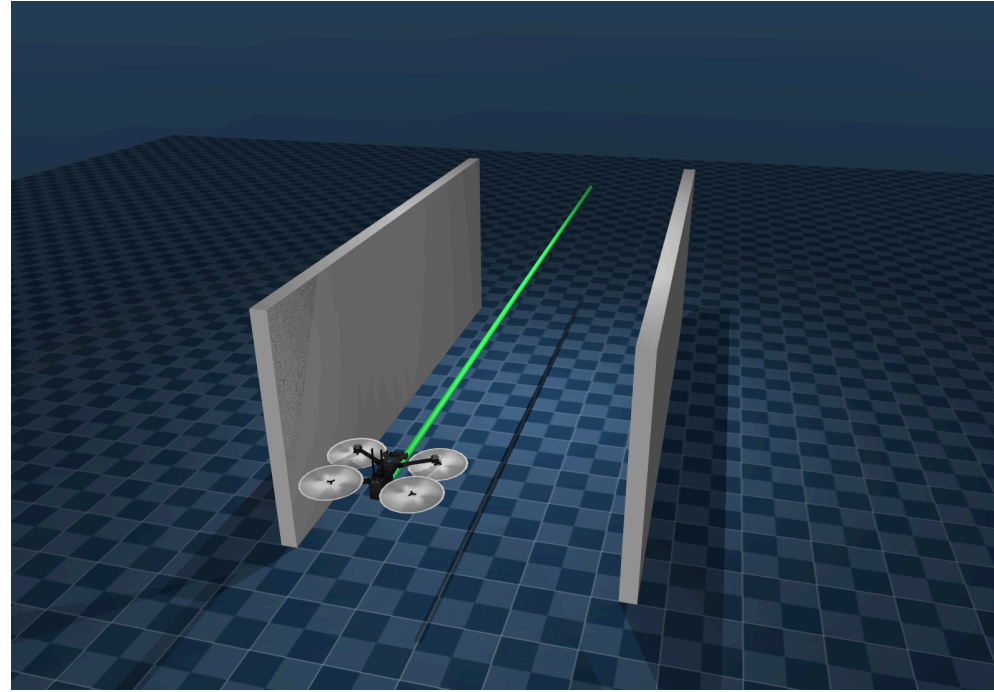
- Endpoint projection keeps more contexts violation-free than per-step projection in seven of nine comparisons.
- The constraint is paid for in distance: no context ends within the 1.8 cm of the task's own test.

Quadrotor: three scenes that ask three different questions



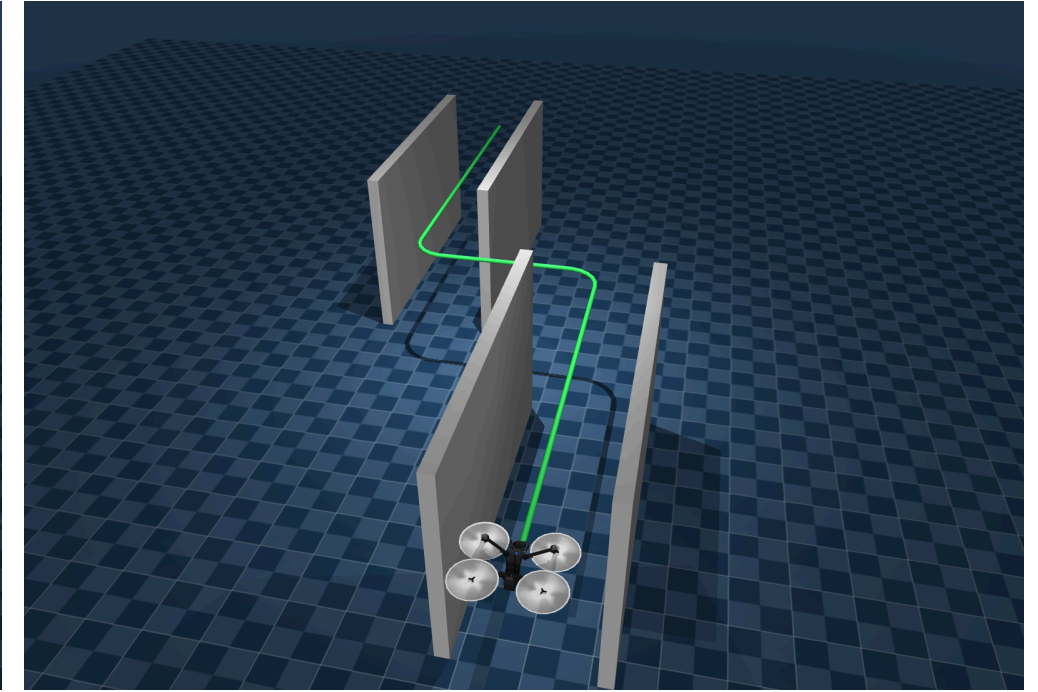
UAV-pillars

D3IL-avoiding flown by the quadrotor



UAV-corridor

constraints that leave the horizontal plane



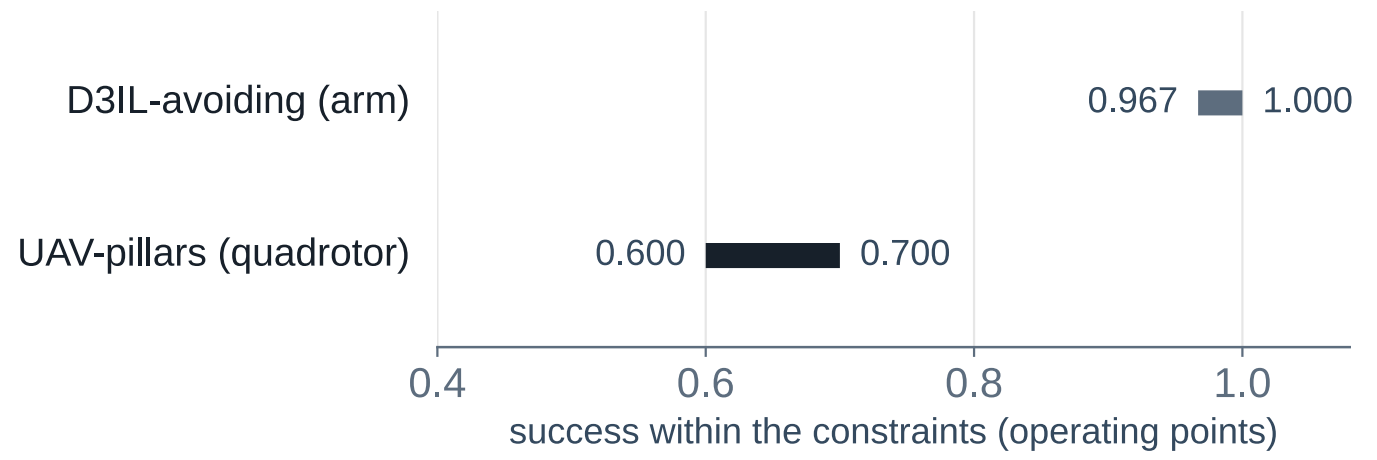
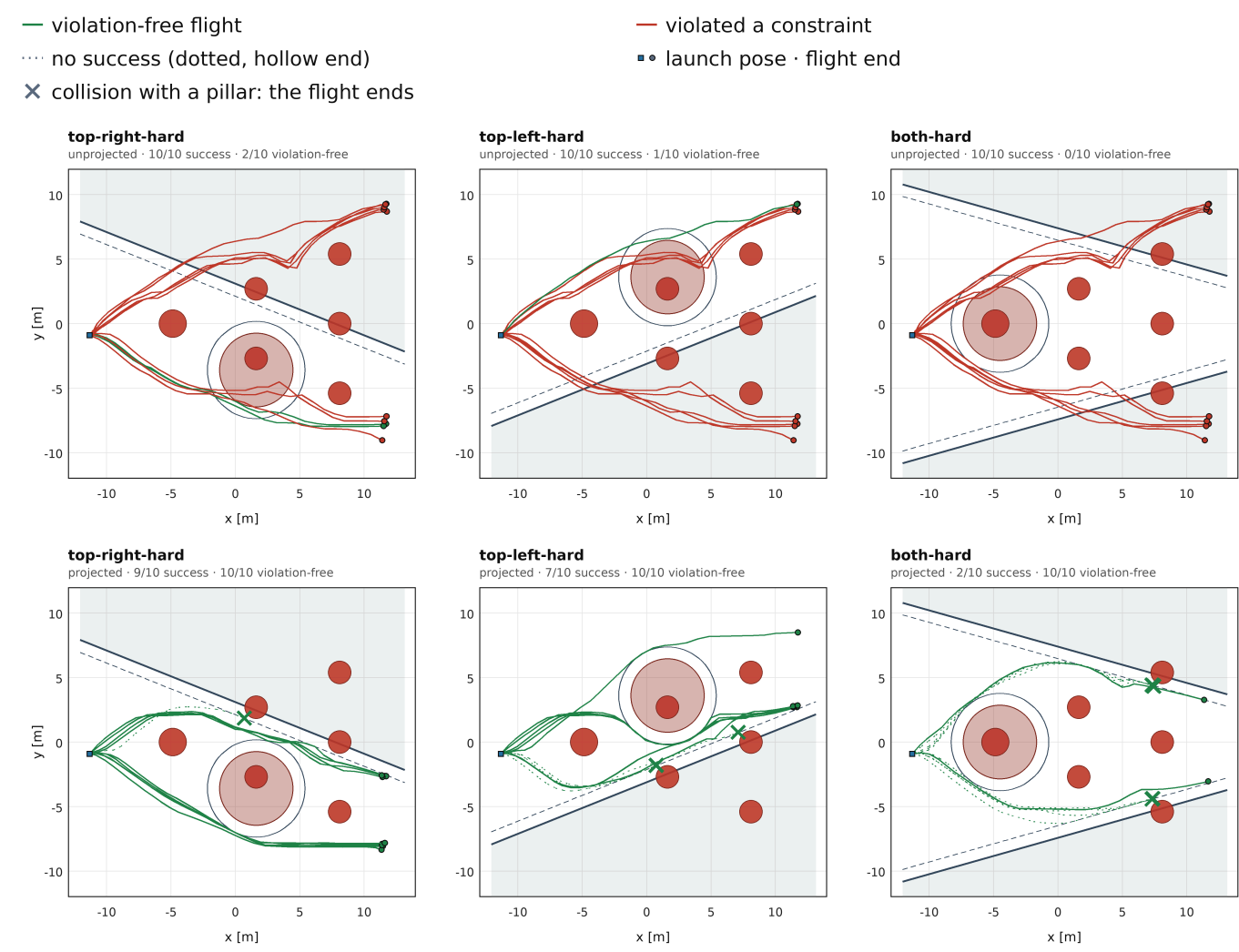
UAV-s-curve

two sharp turns for the tracking controller

Skydio X2 of MuJoCo Menagerie, commanded through its rotor thrusts by a cascaded geometric tracking controller.

MuJoCo flight demos pending.

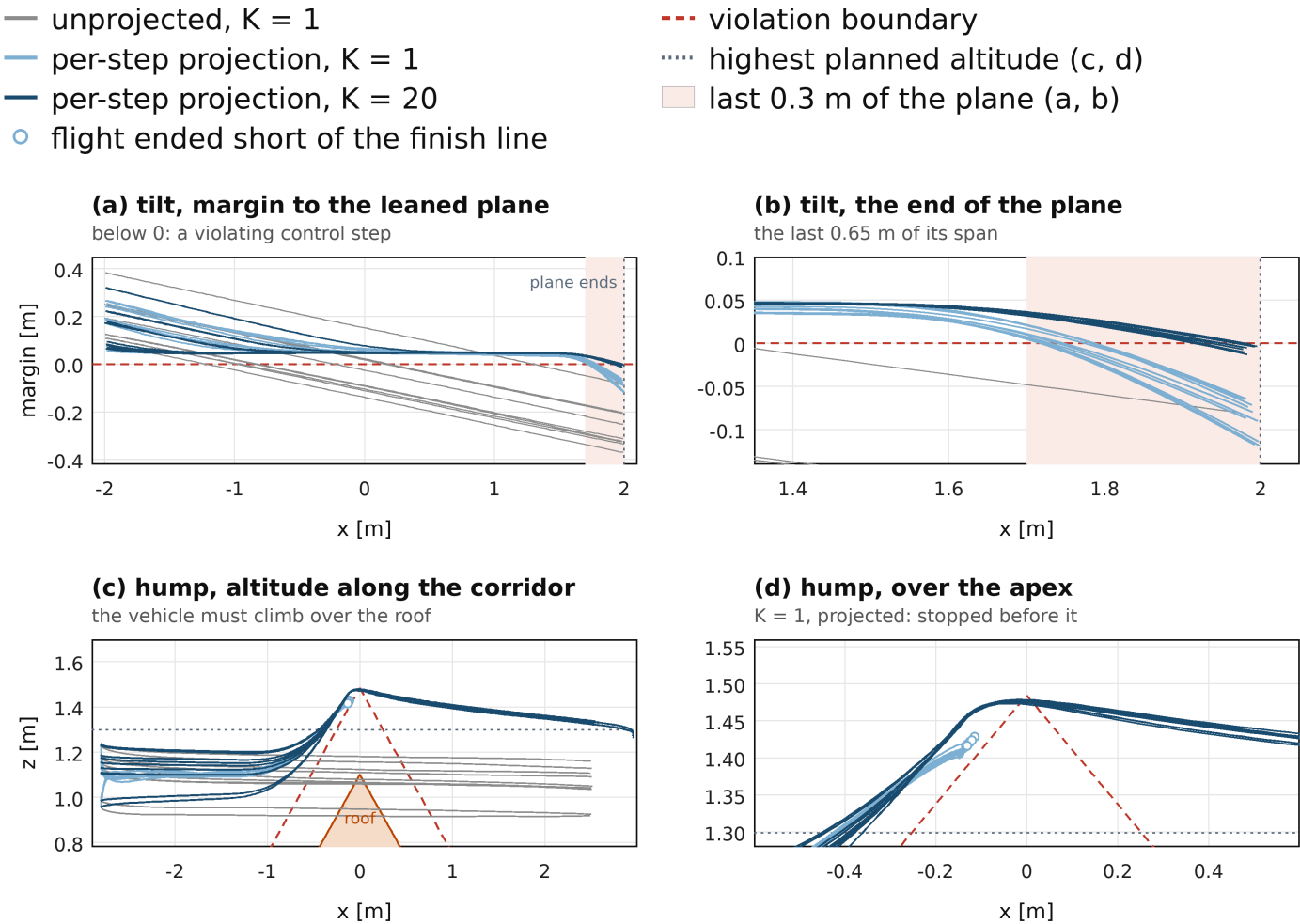
UAV-pillars: the result of D3IL-avoiding holds when its plans are flown



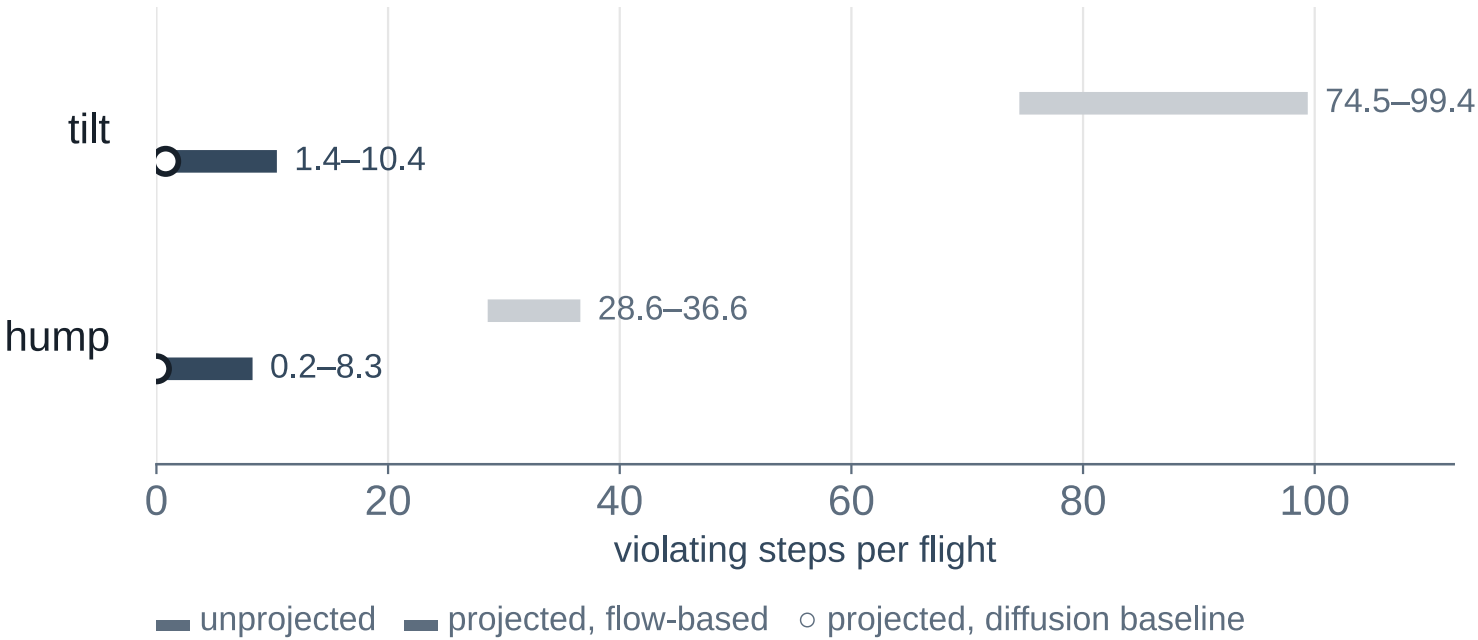
- Success within the constraints at the operating points.
- After projection no flight violates the declared constraints; the vehicle tracks more closely than the arm.
 - Every failure is a collision with one of the five pillars the constraint sets leave out.

Flown paths, unprojected (top) and projected (bottom), three geometries (thesis figure).

UAV-corridor: the projection scheme of DPCC carries over to three dimensions



Margin and altitude along the corridor: unprojected, per-step projection at $K = 1$ and $K = 20$ (thesis figure).

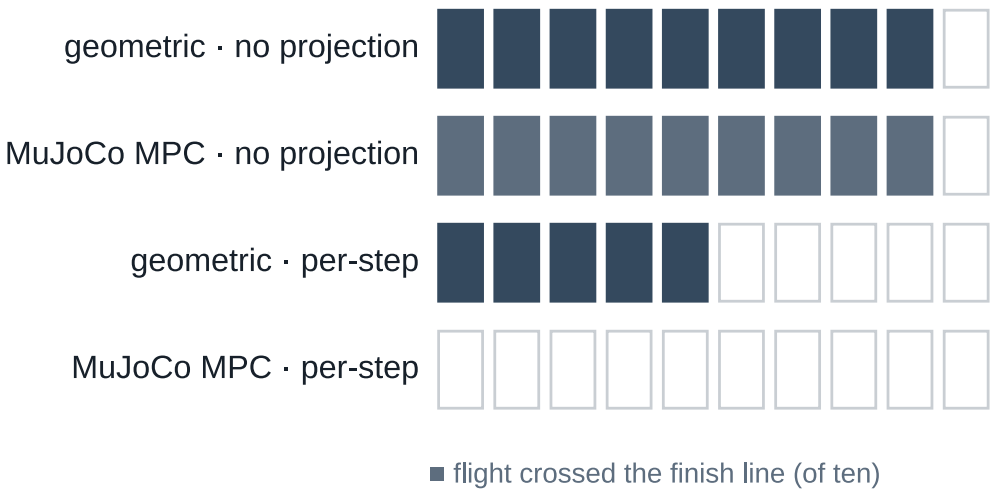
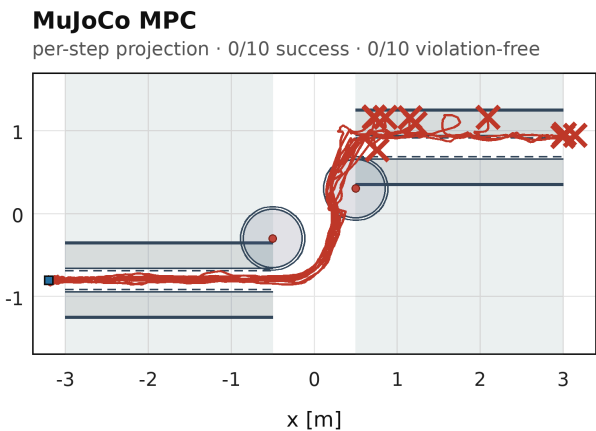
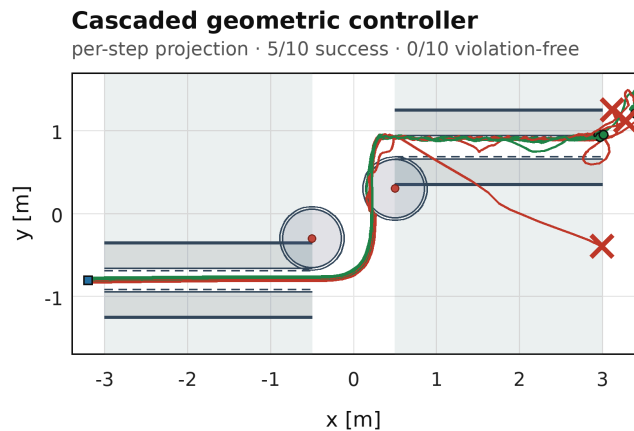
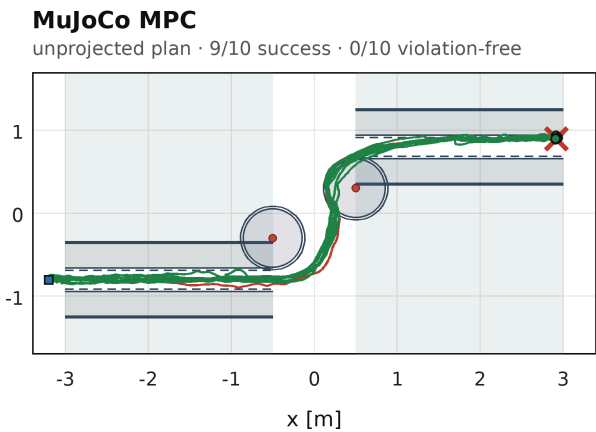
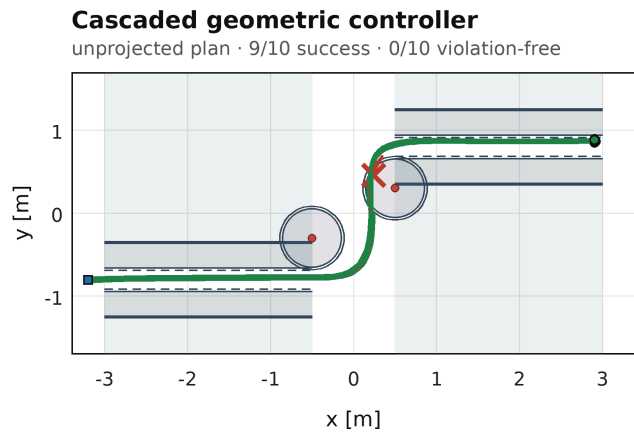


Violating steps per flight.

- Projected, the flights descend under the tilt (540 of 560) and climb over the hump (all 560).
- The objectives cannot be told apart, the budget can; between the two projectors a trade-off.

UAV-s-curve: the controller decides whether a plan is flown

- success: crossed the finish line
- launch pose · flight end
- no success
- × collided or lost control



The configuration of record, by controller and projection.



at one evaluation, no projection, cascaded geometric controller

Cascaded geometric controller (left) and MuJoCo MPC (right); unprojected (top) and per-step projection (bottom) (thesis figure).

By objective.

Three findings across the environments

1. The time a projection takes depends on how far from feasible the plan it receives is.
2. A task saturated at one evaluation hides what a task that rewards more evaluations exposes.
3. The demonstrations bound what a model can show.

Answers to the research questions

- **RQ1 — Yes, on the benchmark of DPCC.** With human demonstrations the average-velocity models come first; with the generated quadrotor demonstrations the objectives tie, or the simplest leads.
- **RQ2 — Where it has steps to guide.** Endpoint projection guides the sample only where the budget and the threshold leave a step before the terminal one; there it is the better projector on the constraint at a comparable planning time. It does not shorten the task, and on the corridor it trades.
- **RQ3 — Where the demonstrations are of the same kind.** The answers hold on camera observations and contact, and the constraints transfer to the quadrotor; the plant and the demonstrations change the rest.

Limitations and outlook

Limitations

- The plan is a sequence of positions; below it is the plant's controller.
- A constraint is what is declared: the projector protects nothing else.
- Every environment is a simulation.

Outlook

- Human demonstrations for the quadrotor scenes.
- A longer planning horizon than eight steps.
- Constraints from perception: depth or range sensing that emits the projector's primitives.

Thank you

Questions